

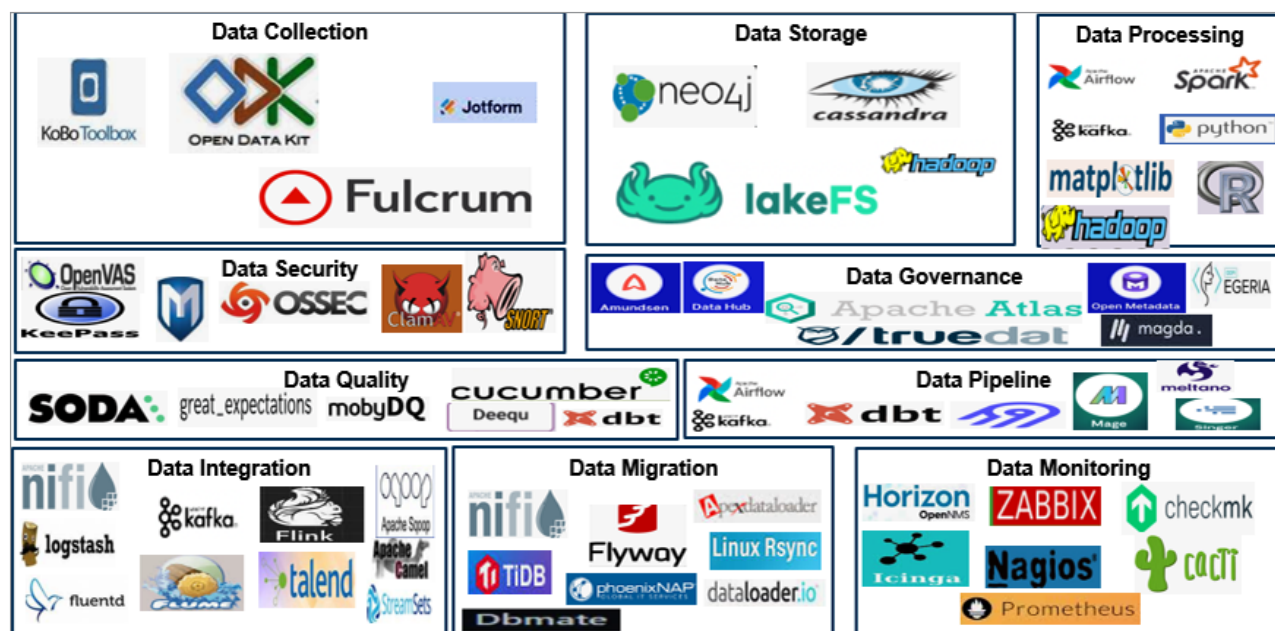


Figure 2: Open source data management ecosystem

Data migration: This is a one-time process of moving data from one database to another. This usually happens because an enterprise is adding a new system or data location. Migration can also mean shifting data formats or applications.

For example, if an enterprise is changing to a new CRM, it needs to migrate data from the current platform to the new one.

Well-known open source based data migration tools are rsync, phinx, Dbmate, Apache NiFi, Flyway and TiDB. These tools help in simple file transfers as well as complex database migrations. They provide scalability, ease of use and active community support.

Data security: Establishing proper security protocols helps to reduce the risk of data breaches. It also safeguards sensitive information. Encryption, access controls, authentication, and intrusion detection systems help protect data across the life cycle.

Open source data security tools that are widely used include Metasploit, OSSEC, OpenVAS, Snort, KeePass, and ClamAV. These tools

can be integrated into various security strategies to protect a wide range of cyber threats.

Data governance: This covers automation of policies, guidelines, principles, and standards for managing data assets. It ensures data quality, accuracy, and compliance with regulatory requirements, building trust in the data. Data governance must be aligned at the enterprise level to realise the business objectives.

Some of the open source data governance tools are Amundsen, DataHub, Apache Atlas, Magda, Open Metadata, Egeria, and TrueDat. These tools offer features like metadata management, data cataloguing

and collaboration to manage data assets effectively.

Data monitoring: This relates to monitoring the quality and relevance of the data, and updating it to improve the performance of applications.

Some of the best open source data monitoring tools are Prometheus, Zabbix, Nagios Core, Checkmk, Cacti, OpenNMS Horizon, Icinga, etc. These tools provide robustness, scalability and active community support.

Having good ways to manage data is important for businesses today. By implementing data management best practices, enterprises can ensure that they are managing their data well enough to maximise its value. **END** 🐧

Acknowledgements

The author would like to thank Santosh Shinde of BTIS, Enterprise Architecture division of HCL Technologies Ltd, for giving the required time and support to help write this article as part of Architecture Practice efforts.

By: Dr Gopala Krishna Behara

The author is an enterprise architect in the BTIS Enterprise Architecture division of HCL Technologies Ltd. He has a total of 28 years of experience in the IT industry.

Disclaimer: The views expressed in this article are that of the author and HCL does not subscribe to the substance, veracity, or truthfulness of the said opinion.